- A H M E D N A B I L A T W A -

Lead/Senior AI Engineer | Data Scientist | Quantitative Researcher
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Lead/Senior AI Engineer and Data Scientist with advanced expertise in **Generative AI, Agentic Frameworks, and Quantitative Finance**. Proven track record of architecting scalable RAG systems and probabilistic machine learning models that drive operational efficiency and revenue growth. Bridging the gap between rigorous academic research (Generative Feature Extraction, RL) and production-grade software engineering (GCP, MLOps). Currently architecting autonomous AI agents for the hospitality and fintech sectors.

TECHNICAL SKILLS

- **GenAI & LLMs:** LangChain, LangGraph, RAG Systems, Haystack, Qdrant (Vector DB), Hugging Face, LLMs APIs.
- **Machine Learning:** PyTorch, JAX, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, Statsmodels.
- **MLOps:** GCP (Cloud Run, BigQuery, Cloud Build), SQL, Docker, Git, Dask.

PROFESSIONAL EXPERIENCE

Prometheus iQ, Contract, Austin, TX, Remote • *Lead AI Engineer* **Jul/2025 - Present**

- **Agentic AI Architecture:** Spearheaded the development of an autonomous Agentic AI framework powering the company's revenue-management engine, enabling automated decisioning across hospitality workflows.
- **Cloud Scalability:** Engineered cloud-native backend systems on **GCP (Cloud Run, BigQuery)**, improving processing efficiency by **50%** and enabling real-time insight delivery.
- **DevOps:** Implemented CI/CD pipelines (Cloud Build) to accelerate deployment speed and support rapid product iteration.

Finetooth Analytics, Contract, N.Y., U.S, Remote • *Quant AI Researcher/Engineer* **Apr/2025 – May/2025**

- **LLM Integration:** Architected high-availability API pipelines integrating LLMs (**DeepSeek-V3, GPT-4o, Grok-3**) with financial data providers (e.g., Alpaca) for real-time market analysis.
- **Alpha Generation:** Architected quantitative models bridging U.S. equity markets and real estate analytics, creating novel cross-domain signals for risk mitigation and alpha generation.

CCIT, AAST, Part-time, Alexandria, Egypt • *Lecturer* **Sep/2022 - Present**

- Led the enhancement of Information Systems & Big Data Analytics curricula, integrating data-driven content that increased student engagement by 25% based on user interaction analysis.
- Developed data-driven teaching methodologies, analyzing classroom metrics to enhance instructional effectiveness, significantly.

Upwork / Independent Consultant, Remote • *Data Scientist | AI Research Engineer* **Jul/2021 - Present**

- Delivered specialized machine learning solutions for international clients, focusing on time-series forecasting and financial modeling.
- **iSmile Technologies (Internship Component):** Developed an automated time-series ML system for stock movement prediction, significantly enhancing forecasting accuracy (Aug 2021 – Jan 2022).

Beauty On-Demand / Sephora-Toi, Full-time, Cairo, Egypt • *Business Development* **Oct/ 2016 - May/2021**

- **Beauty On-Demand** • *Business Development Manager* • **Mar/ 2018 - May/2021**
  - Designed and managed a product database with over 1,000 entries, leveraging data structuring to boost operational efficiency and improve product availability by 20%.
  - Processed and optimized high-quality product images for over 11,000 items using Excel, VBA, and Photoshop, enhancing online shopping metrics and increasing customer satisfaction by 25%.
- **Sephora-Toi** • *Business Information Systems* • **Oct / 2016 - Feb/2018**
  - Created over 2,000 digital products, expanding the company's product portfolio and contributing to a 30% increase in digital sales

OPEN-SOURCE PROJECTS

- **Generative Models for Feature Extraction in JAX**: Implemented stochastic feature extraction techniques for high-dimensional data. [Link to Repo]
- **JPX Ranker by Tokyo Stock Exchange (Kaggle Competition):** Ranking system implementing advanced financial ML principles with rigorous statistical validation, data governance, and robust out-of-sample estimation. [Link to Repo]
- **Lumibot (HFT & Advanced ML Fork):** Developed a robust fork of the Lumibot algorithmic trading framework. Engineered High-Frequency Trading (HFT) capabilities with **Alpaca**, optimized API calling architecture, and integrated advanced financial ML adaptability with enhanced backtesting statistics. [Link to Repo]
- **Linguistic Similarity Model (PyTorch):** Fine-tuned DeBERTa-v3 for semantic similarity tasks. [Hugging Face Link]

EDUCATION

Ph.D Degree in Information Systems • Computing and Information Technology, Arab Academy for Science, Technology & Maritime Transport, Abo-Kir, Alexandria, Egypt

Graduation Year: *2027*

Specializing in Reinforcement Learning in Finance, Specialized in High-Risk Regimes.

Master Degree in Information Systems • Computing and Information Technology, Arab Academy for Science, Technology & Maritime Transport, Abo-Kir, Alexandria, Egypt

Graduation Year: *2024* (Oct 2022 — Dec 2024)

Specializing in Probabilistic Feature Extraction and Machine Learning in Finance, Excellent with honors, (GPA 3.83)

Bachelor's Degree in Information Systems (Minor Computer Science) • Computing and Information Technology, Arab Academy for Science, Technology & Maritime Transport, Abo-Kir, Alexandria, Egypt (Sep 2015 — Jul 2019)

Graduated with Excellent with honors (GPA 3.44)

CERTIFICATES & COURSES

Google Cloud, Google, *Professional Machine Learning Engineer* — Sep/2021 – Jan/2022

edX IBM, edX, *Python Data Science* — Sep/2020 – May/2021

EMC Education Services, Dell, *Academic Associate, Data Science and Big Data Analytics* — Sep/2018 – Jan/2019

CONFERENCES & PUBLICATIONS

- A. N. Atwa, M. Kholief and A. Sedky, "Probabilistic Feature Extraction in Portfolio Optimization: Integrating PPCA with Mean Reversion and Mean-Variance Strategies," *2024 34th International Conference on Computer Theory and Applications (ICCTA)*, Alexandria, Egypt, 2024. https://doi.org/10.1109/ICCTA64612.2024.10974778
- Nabil Atwa, A., Sedky, A. & Kholief, M. Challenges of conventional feature extraction techniques for multivariate time-bars assets. *Int J Data Sci Anal* 20, 6333–6355 (2025). https://doi.org/10.1007/s41060-025-00824-w
- Atwa, A.N., Kholief, M. and Sedky, A., 2026. Generative modeling for high-dimensional sparse data: Probabilistic feature extraction in high-risk financial regimes. *Engineering Applications of Artificial Intelligence*, *164*, p.113376. https://doi.org/10.1016/j.engappai.2025.113376.

LANGUAGES & FRAMEWORKS

Python • SQL • NumPy • SciPy • Statsmodels • SymPy • Pandas • Dask • Matplotlib • Seaborn • Plotly • NLTK • SpaCy • PyCaret • scikit-learn • XGBoost • LightGBM • CatBoost • nonconformist • MAPIE • crepes • aws-fortuna • NeuralProphet • statsforecast • Skfolio • OpenML • TensorFlow • PyTorch • JAX • Flax • Haiku • Optax • Chex • Distrax • PyMC3 • RLlib • Stable-Baselines3 • Gymnasium • MuJoCo • Brax • ChainerRL • Acme